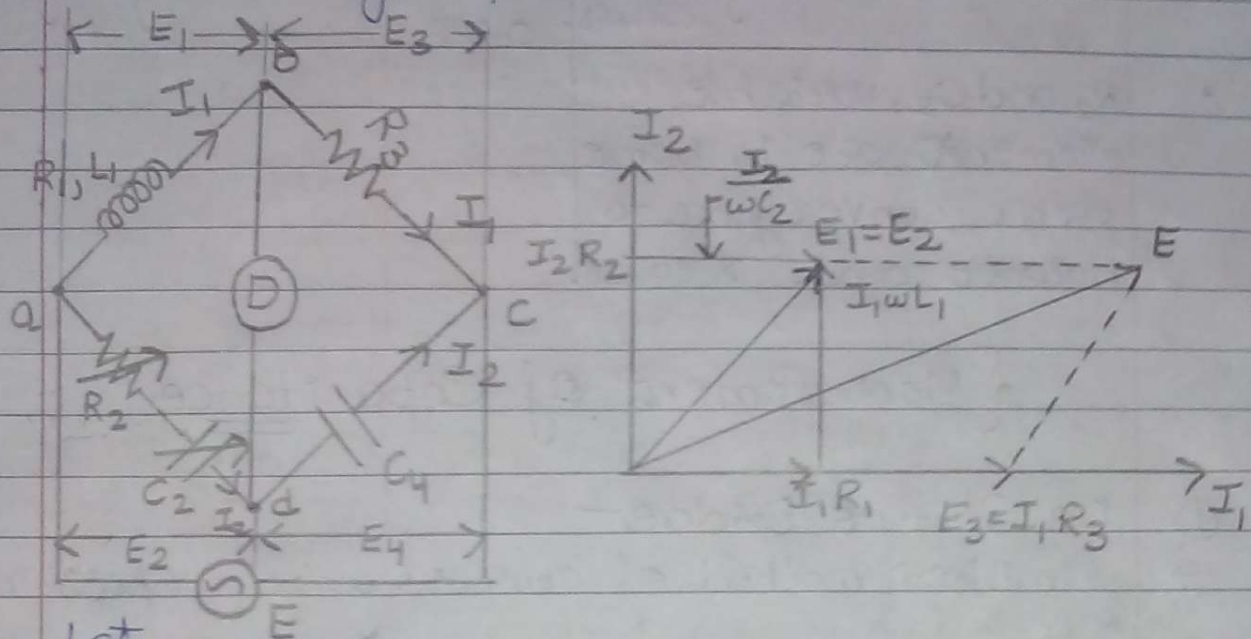


Owen's Bridge :-

This bridge is used for measurement of an inductance in terms of capacitance.



Let

- L_1 = unknown self inductance of resistance R_1
- R_2 = variable non-inductive resistance
- R_3 = fixed non-inductive resistance
- C_2 = variable standard capacitor
- and C_4 = fixed standard capacitor

At balance,

$$(R_1 + j\omega L_1) \left(\frac{1}{j\omega C_4} \right) = \left(R_2 + \frac{1}{j\omega C_2} \right) R_3$$

Separating real and imaginary parts,

$$L_1 = R_2 R_3 C_4$$

$$R_1 = R_3 \frac{C_4}{C_2}$$

Advantages

- Balance eq. are simple and do not have frequency component

Disadvantages

- Bridge requires variable capacitor which is expensive item.

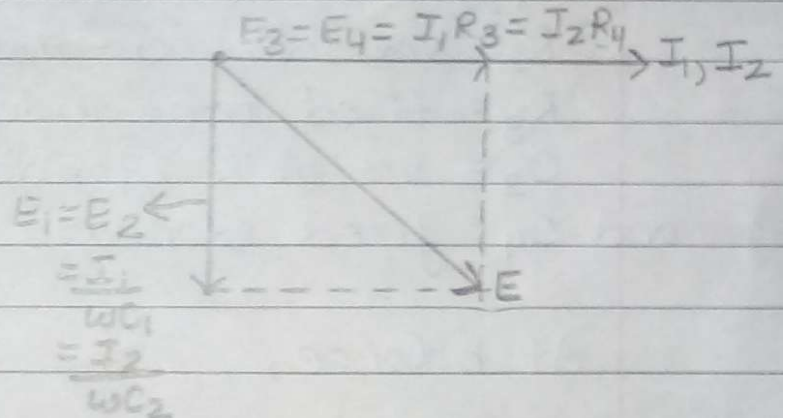
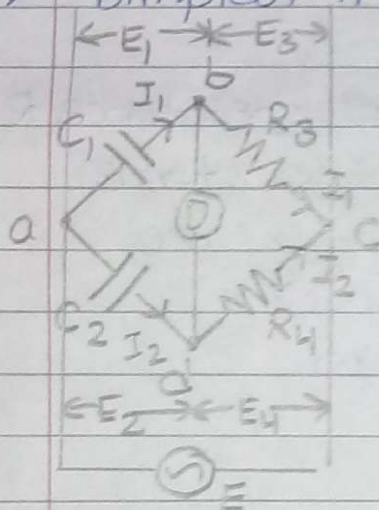
- Bridge can be used over wide range of measurement of inductances.
- R_2 and C_2 , variable elements are in same arm, convergence to balance cond. is easier.

Value of capacitance C_2 tends to become large when measuring high Q coils.

{ Measurement of Capacitance }

① De Sauty's Bridge—

→ Simplest method of comparing two capacitances.



At balance,

$$\left(\frac{1}{j\omega C_1} \right) R_4 = \left(\frac{1}{j\omega C_2} \right) R_3$$

$$\text{or } C_1 = \frac{C_2 R_4}{R_3}$$

- Balance can be obtained by varying either R_3 or R_4 .

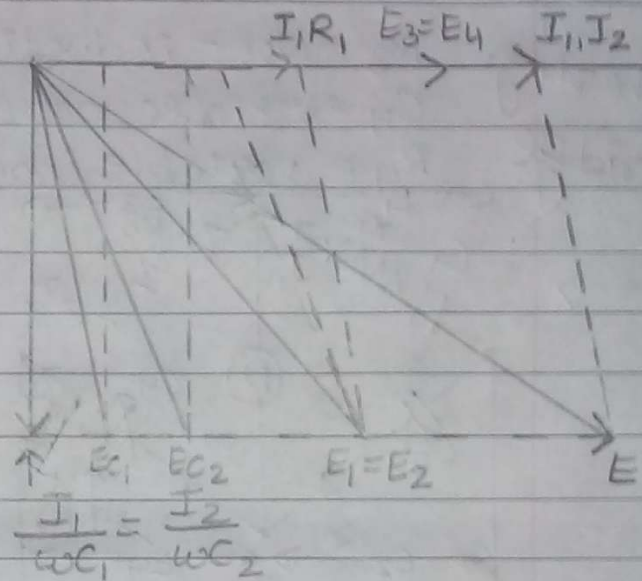
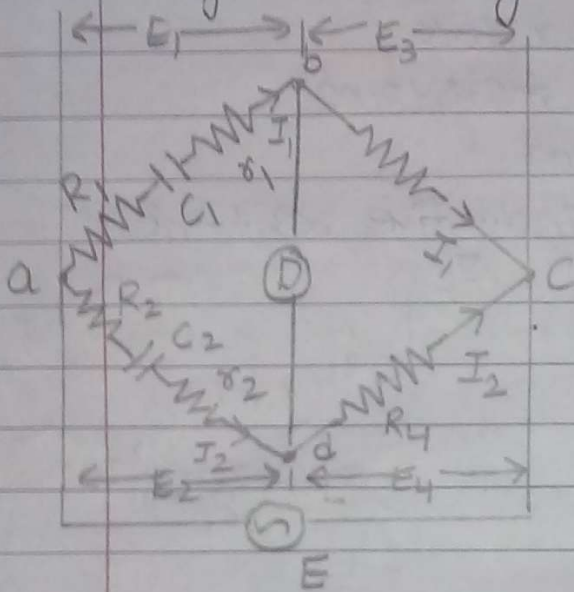
→ advantage is simplicity

→ Disadvantage :-

- only loss-less capacitor like air capacitor are compared.

- It is impossible to find when both capacitors are not free from dielectric loss.

Modified De Sauty's Bridge:-



At balance,

$$\left(R_1 + \delta_1 + \frac{1}{j\omega C_1}\right) R_4 = \left(R_2 + \delta_2 + \frac{1}{j\omega C_2}\right) R_3$$

hence, $\frac{C_1}{C_2} = \frac{R_2 + \delta_2}{R_1 + \delta_1} = \frac{R_4}{R_3}$ — (1)

Dissipation factors for capacitors are -

① $D_1 = \tan \delta_1 = \omega C_1 \delta_1$ ② $D_2 = \tan \delta_2 = \omega C_2 \delta_2$

We know, $\frac{C_1}{C_2} = \frac{R_2 + \delta_2}{R_1 + \delta_1}$ (from 1)

$$C_2 \delta_2 - C_1 \delta_1 = C_1 R_1 - C_2 R_2$$

$$\omega C_2 \delta_2 - \omega C_1 \delta_1 = \omega (C_1 R_1 - C_2 R_2)$$

$$D_2 - D_1 = \omega (C_1 R_1 - C_2 R_2) \quad \text{--- (2)}$$

But $C_1/C_2 = R_4/R_3 \rightarrow C_1 = C_2 R_4/R_3$

$$D_2 - D_1 = \omega C_2 \left[\frac{R_4 R_1 - R_2}{R_3} \right]$$

If dissipation factors of one of capacitors is known, the other can be determined.

- Method does not give accurate result.
- Quantities are very large.

Schering Bridge -

Let, C_1 = capacitor whose capacitance is to be found

δ_1 = series resistance represent loss in C_1

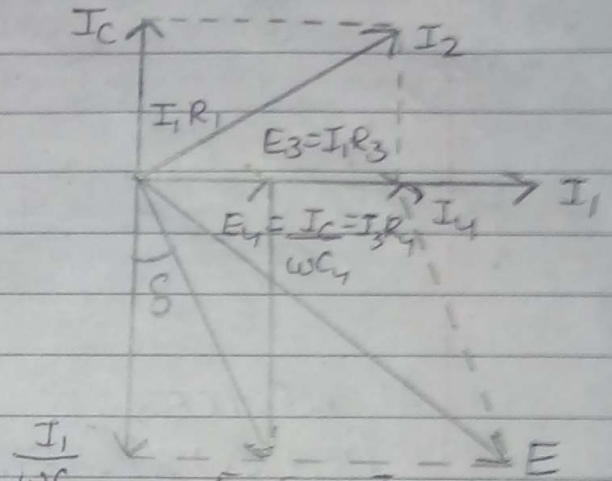
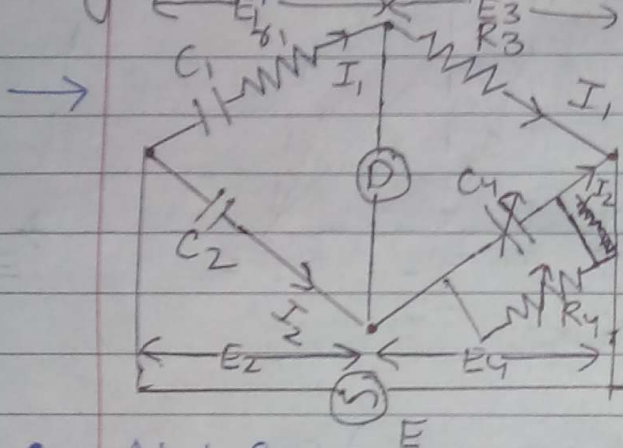
C_2 = standard capacitor

R_3 = Non-inductive resistance

C_4 = variable capacitor

R_4 = variable non-inductive resistance

Low Voltage Schering Bridge



- At balance,

$$\left(\delta_1 + \frac{1}{j\omega C_1} \right) \left(\frac{R_4}{1 + j\omega C_4 R_4} \right) = \frac{1}{j\omega C_2} \quad E_1 = E_2 = \frac{I_2}{\omega C_2}$$

$$\left(\delta_1 + \frac{1}{j\omega C_1} \right) R_4 = \frac{R_3}{j\omega C_2} (1 + j\omega C_4 R_4)$$

$$\frac{\delta_1 R_4}{\omega C_1} - \frac{j R_4}{\omega C_1} = - \frac{j R_3}{\omega C_2} + \frac{R_4 R_3 C_4}{C_2}$$

Equating real & imaginary,

$$\delta_1 = \frac{R_3 C_4}{C_2} \quad \text{and} \quad C_1 = \frac{C_2 R_4}{R_3}$$

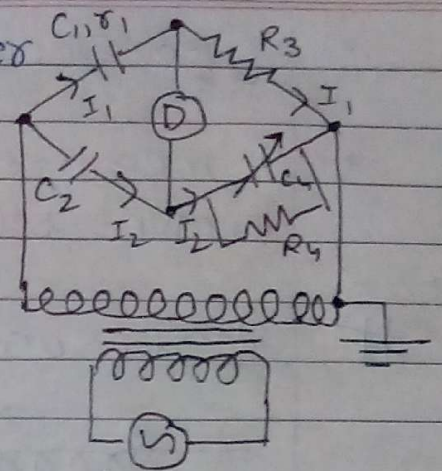
Dissipation factor

$$D_1 = \tan \delta = \omega C_1 \delta_1 = \omega \left(\frac{C_2 R_4}{R_3} \times \frac{R_3 R_4}{C_2} \right)$$

$$D_1 = \omega C_4 R_4$$

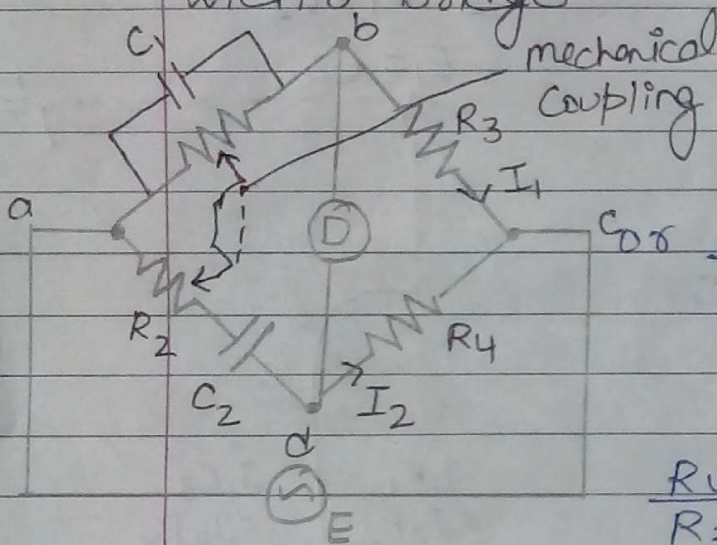
② High Voltage :- (Schering Bridge)

- Detector $\rightarrow$ Vibration galvanometer
- Spark gap connected.
- impedance of arms are large
- Detector is at low potential
- Earthed screens are provided (to avoid errors)



(Measurement of Frequency)

- Wien's Bridge (for frequency)



At balance,

$$R_4 \left(\frac{R_1}{1 + j\omega C_1 R_1} \right) = R_3 \left(\frac{R_2 - \frac{1}{\omega C_2}}{\omega C_2 R_1} \right)$$

or

$$\frac{R_4}{R_3} = \frac{R_2 + \frac{C_1}{C_2}}{R_1} + j \left(\frac{\omega C_1 R_2 - \frac{1}{\omega C_2 R_1}}{\omega C_2 R_1} \right)$$

Equating,

$$\frac{R_4}{R_3} = \frac{R_2 + \frac{C_1}{C_2}}{R_1} \quad \omega C_1 R_2 - \frac{1}{\omega C_2 R_1} = 0$$

from which, $\omega = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}$

and frequency $f = \frac{1}{2\pi \sqrt{R_1 R_2 C_1 C_2}} \text{ Hz}$

- In most cases, $R_1 = R_2 = R$ and $C_1 = C_2 = C$

then $\frac{R_4}{R_3} = 2 \rightarrow \boxed{f = \frac{1}{2\pi RC}}$

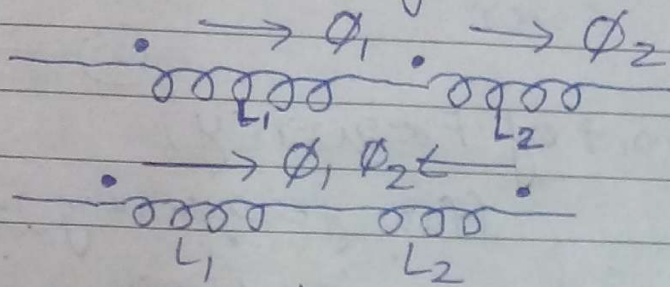
- Applications — ① Audio and HF oscillators
② Harmonic distortion Analyser

Heaviside Bridge:-

(Heaviside Mutual Inductance Bridge)

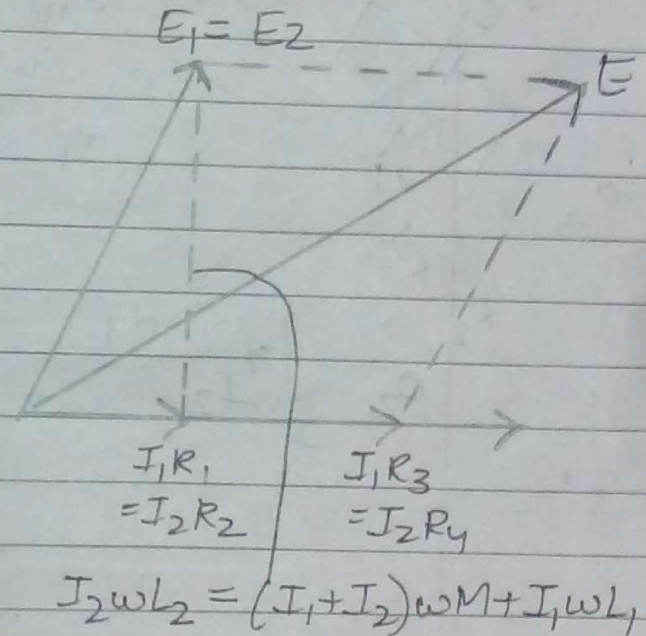
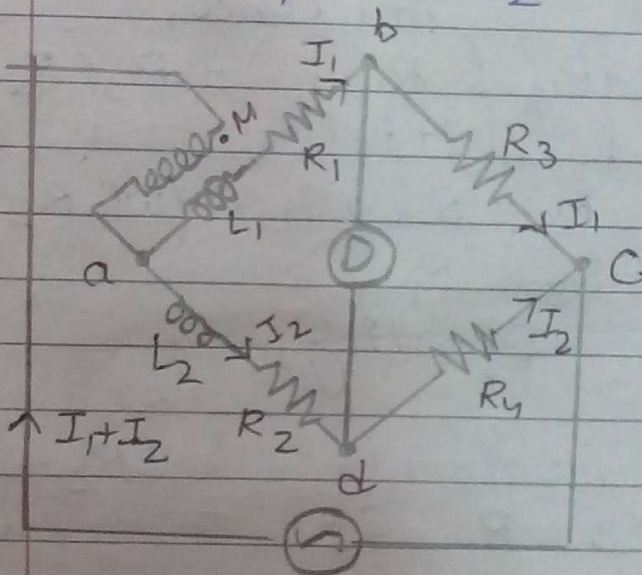
- It measures mutual inductance in terms of known self inductance.

Measurement of Mutual Inductance:-



$$L_{e1} = L_1 + L_2 + 2M$$

$$L_{e2} = L_1 + L_2 - 2M$$



At balance,

$$I_1 R_3 = I_2 R_4$$

and

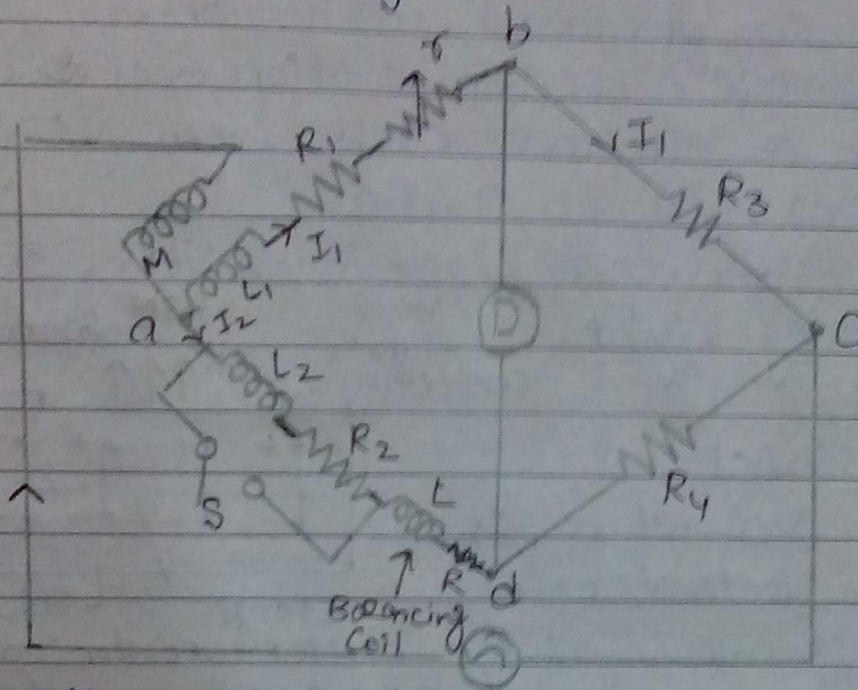
$$(I_1 + I_2) j \omega M + I_1 (R_1 + R_3 + j \omega L_1) = I_2 (R_2 + R_4 + j \omega L_2)$$

Therefore,

$$R_1 = \frac{R_2 R_3}{R_4}$$

$$M = \frac{R_3 L_2 - L_1 R_4}{R_3 + R_4}$$

- Campbell's Modification of Heaviside Bridge:-
- This is used to measure self inductance in terms of mutual inductance.



- With switch open,

$$L_2 + L = \frac{M_1(R_3 + R_4) + R_4 L_1}{R_3} \quad \text{and} \quad R_2 + R = \frac{(R_1 + r_1) R_4}{R_3}$$

- With switch closed,

$$L = \frac{M_2(R_3 + R_4) + R_4 L_1}{R_3} \quad \text{and} \quad R = \frac{(R_1 + r_2) R_4}{R_3}$$

$$\therefore \left[L_2 = (M_1 - M_2) \left(1 + \frac{R_4}{R_3} \right) \right]$$

$$\text{and} \left[R_2 = (r_1 - r_2) \cdot \frac{R_4}{R_3} \right]$$

- Advantage:-
- Method adopted to eliminate the effect of leads.

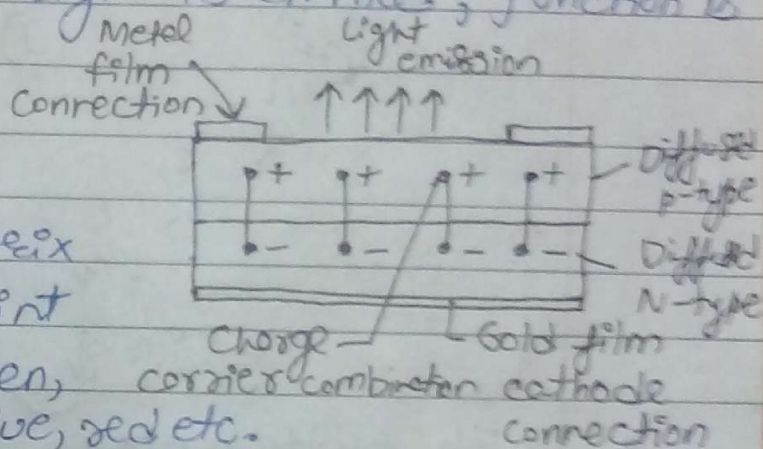
• Various Display Devices are :-

① Light Emitting Diode (LED) —

- PN junction device emit light when current passes through in forward direction.
- Charge carrier recombination occurs at PN junction as electrons cross from N side and recombine holes on P side.
- If it is translucent light is emitted, junction is source of light.

→ Uses —

- ① making (GaAsP)
- ② segment and dot matrix
- ③ to form decimal point
- ④ emits diff colour green, blue, red etc.



② Liquid Crystal Diodes (LCD) —

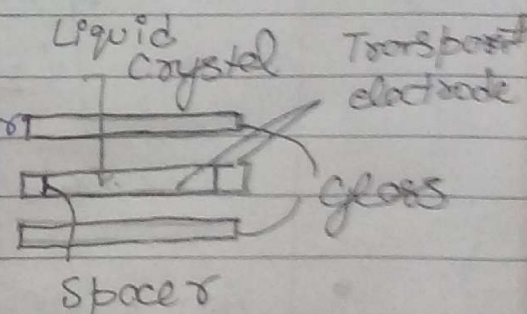
- Two type — Dynamic scattering type.
- field effect type.

- material → organic compound have optical properties of crystal

- Layered b/w glass sheet with transparent electrodes deposited on inside faces.

(Dynamic Scattering)

- When liquid is activated molecular turbulence cause light scattered and cell appears bright.



Liquid crystal cell [Transmissive
reflective